

Evaluating the Robustness of Computer Vision Models

Project by: Ayelet & Noa

BDD100K road scenes · YOLOv8 · Mask R-CNN · MiDaS

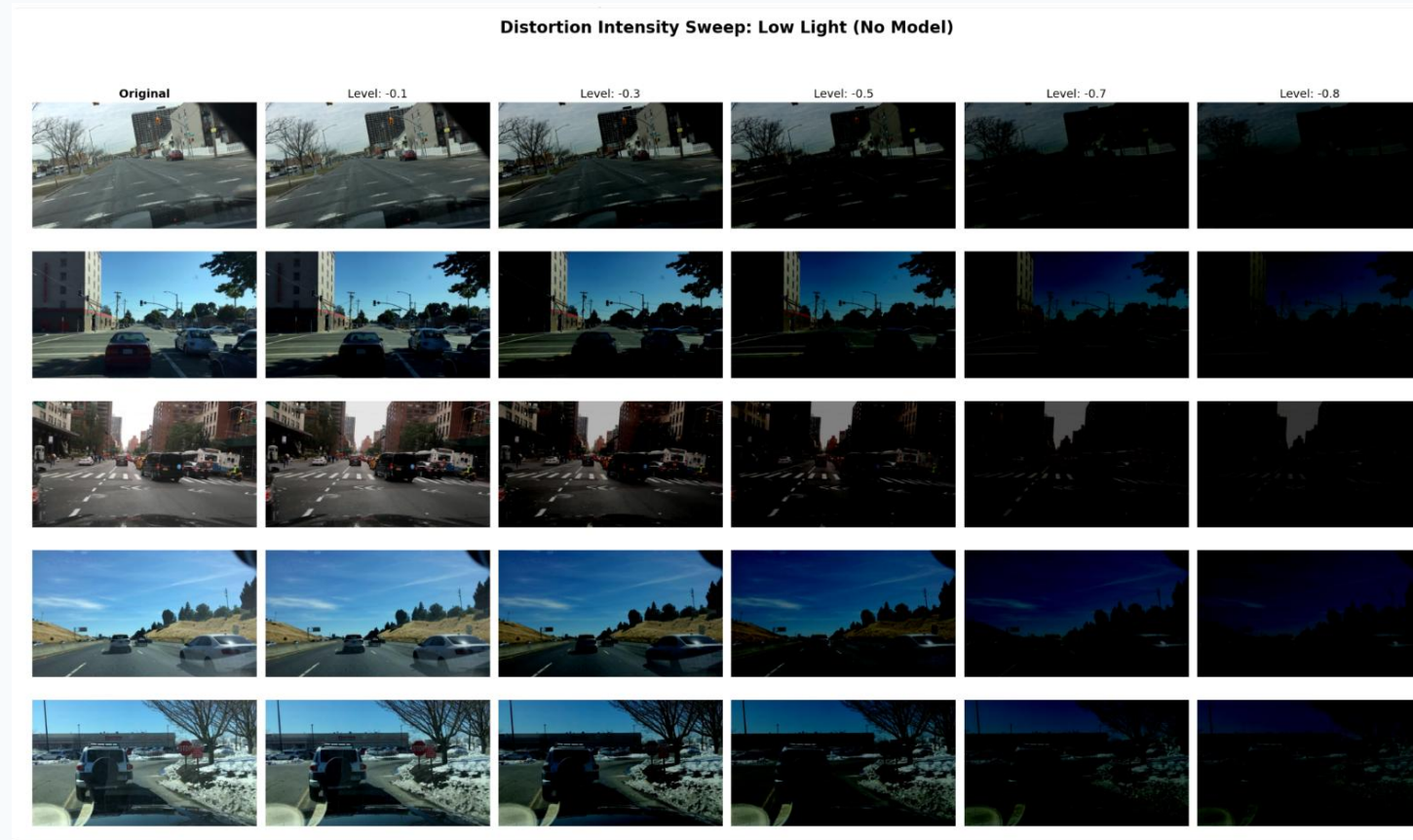
Project Goal

Measure how vision models behave when image quality drops

Test three tasks: object detection, instance segmentation and depth estimation

Compare clean baselines, classical enhancement and fine-tuning

Identify when simple filtering is enough and when model adaptation is needed



Example: gradual low-light degradation

Evaluation Method

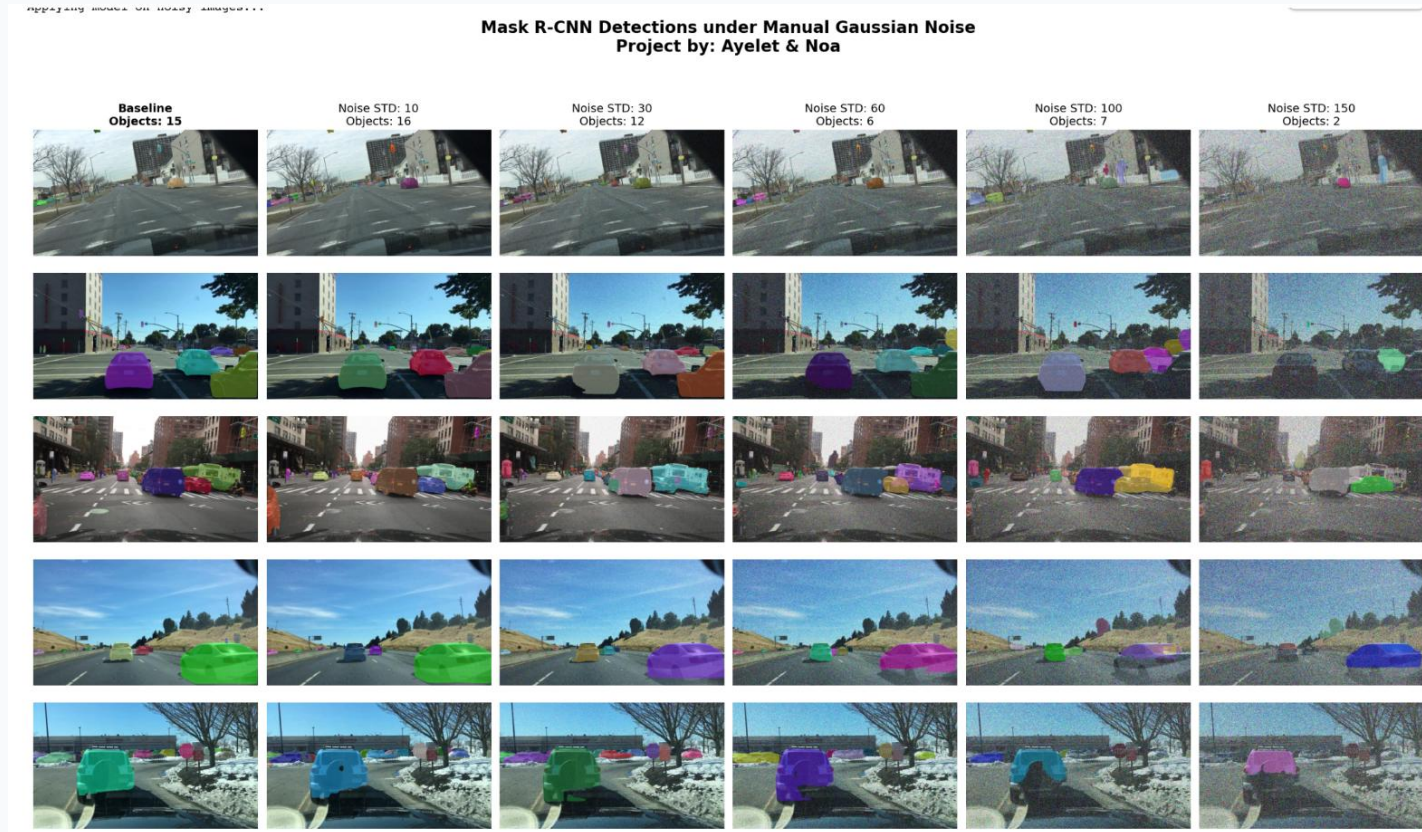
Start from clean images and define a baseline output

Apply one distortion at several severity levels

Run the model again and compare against the clean baseline

Use Recall and Precision (when needed)

for detection and segmentation Use RMSE and SSIM for depth-map comparison



Clean baseline outputs were used as the reference for later comparisons

Models and Metrics

YOLOv8

Object detection

Recall vs. distortion level

Confidence loss and misclassification

Baseline Check: 5 Busy Images vs. YOLOv8 Detections



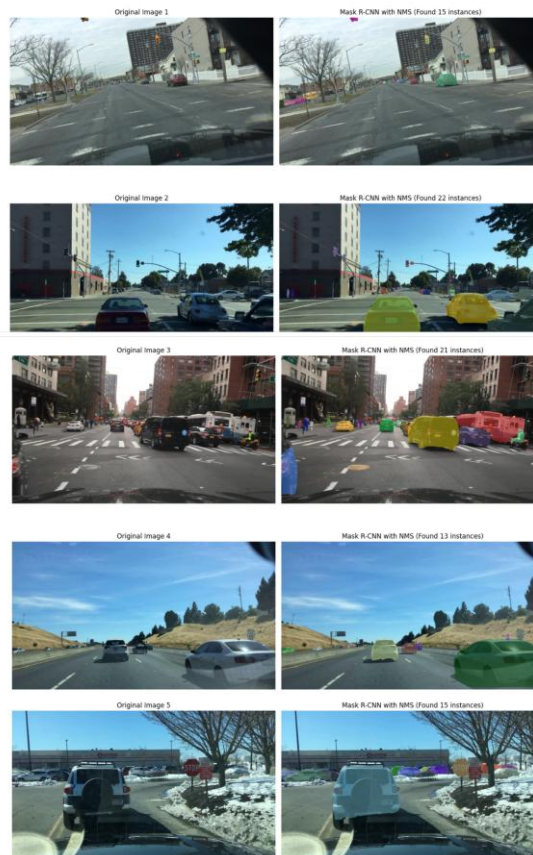
Mask R-CNN

Instance segmentation

Recall and Precision

Effect of masks under noise and compression

Step 1: Final Mask R-CNN Baseline (NMS Enabled)
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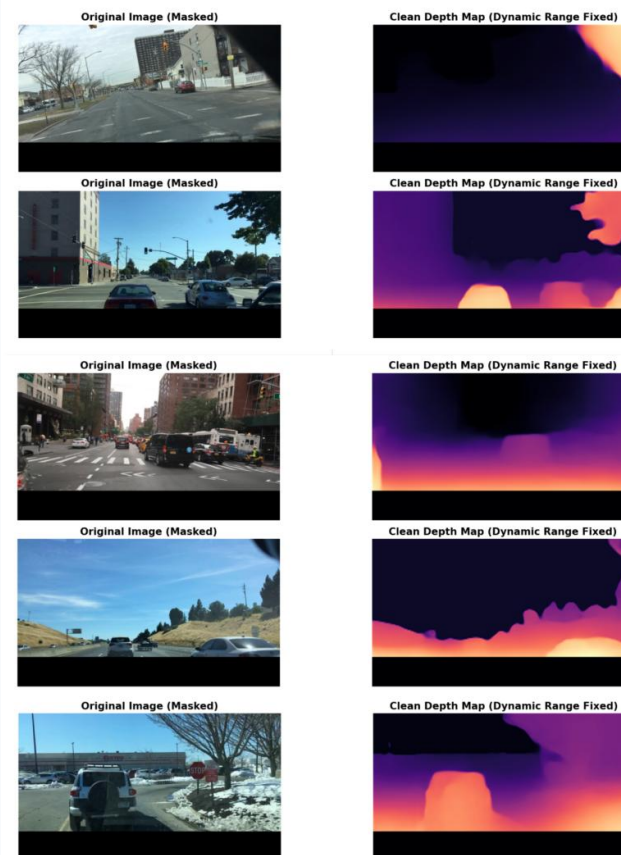
MiDaS

Monocular depth estimation

RMSE, lower is better

SSIM, higher is better

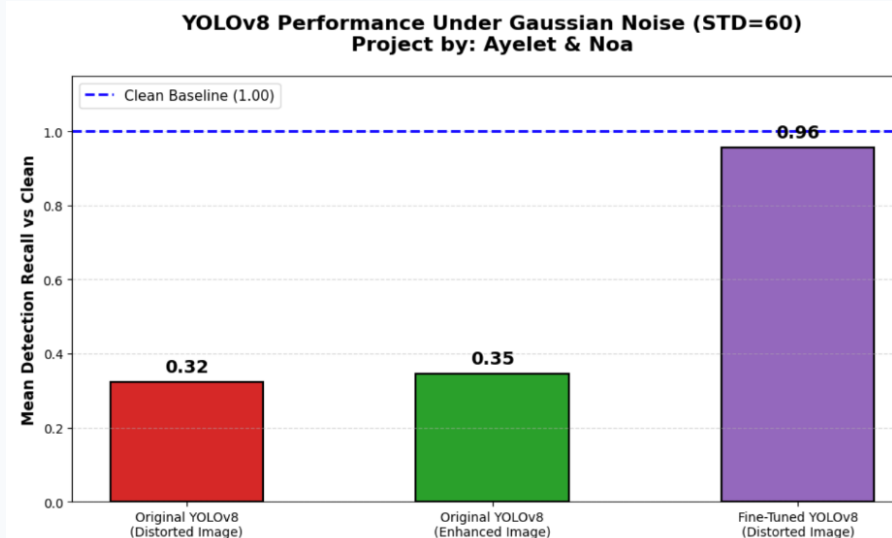
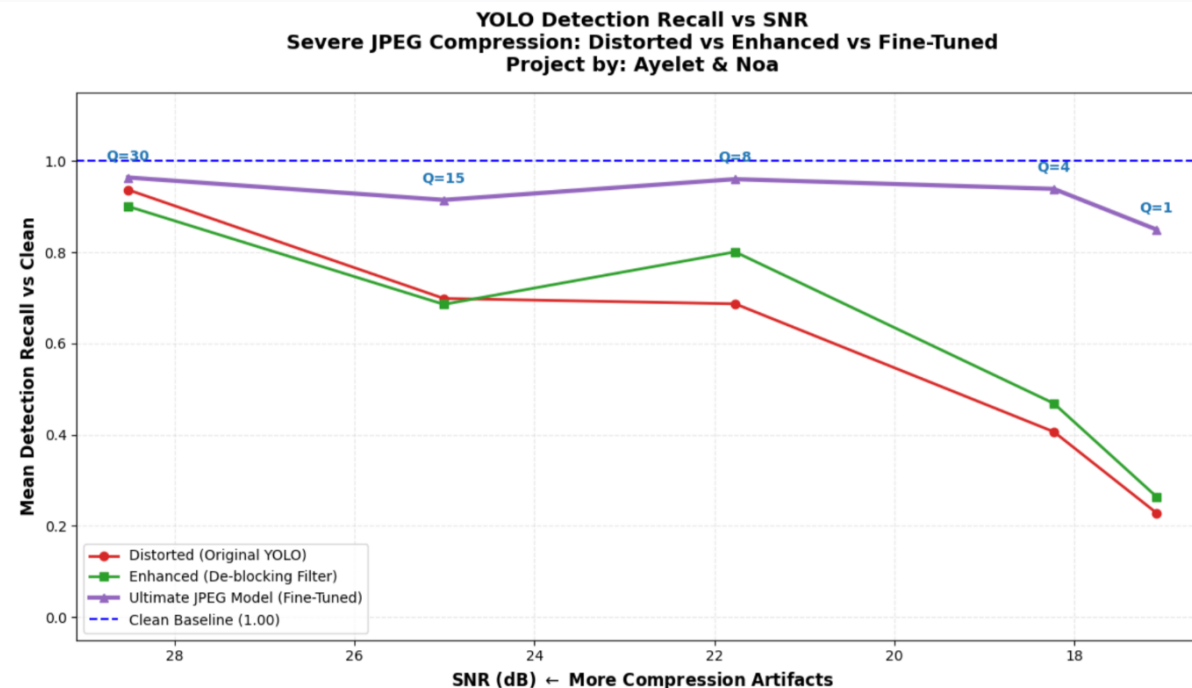
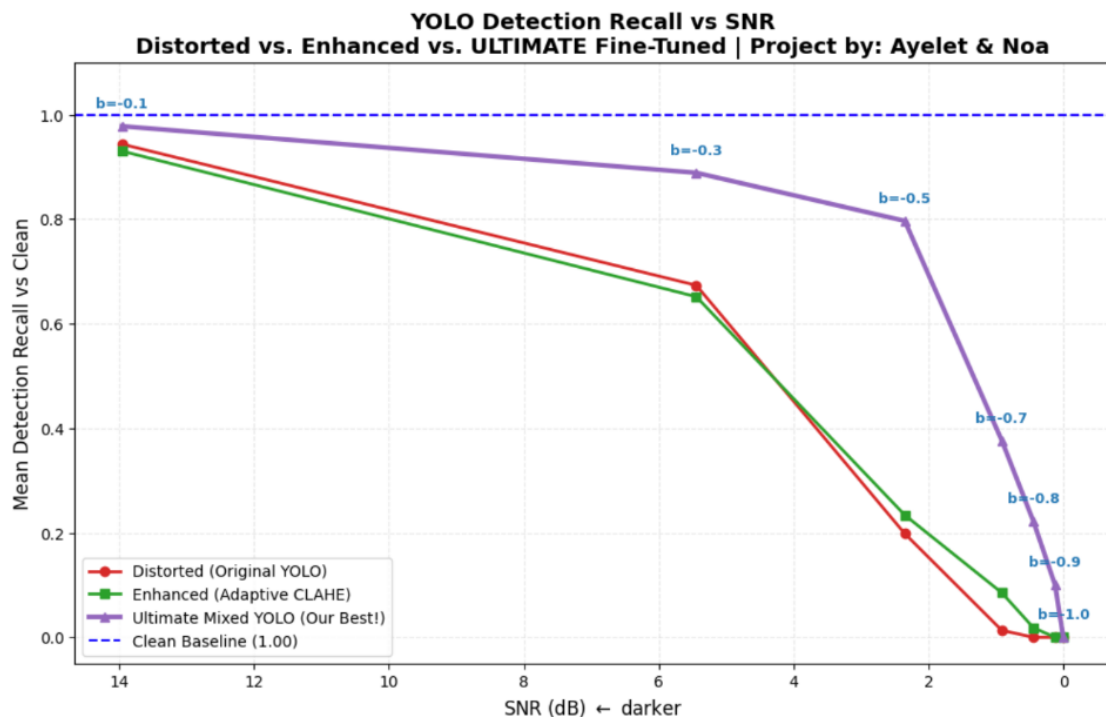
Clean Baseline: Depth Estimation with ROI Fix
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One dataset, three tasks, separate metrics

YOLOv8: Detection Robustness

Low light gradually removes visible objects from the detector
Gaussian noise reduces feature stability and confidence
JPEG artifacts may create false object-like structures
Fine-tuning improves recall under difficult distortions, but the original model can remain best under mild distortion

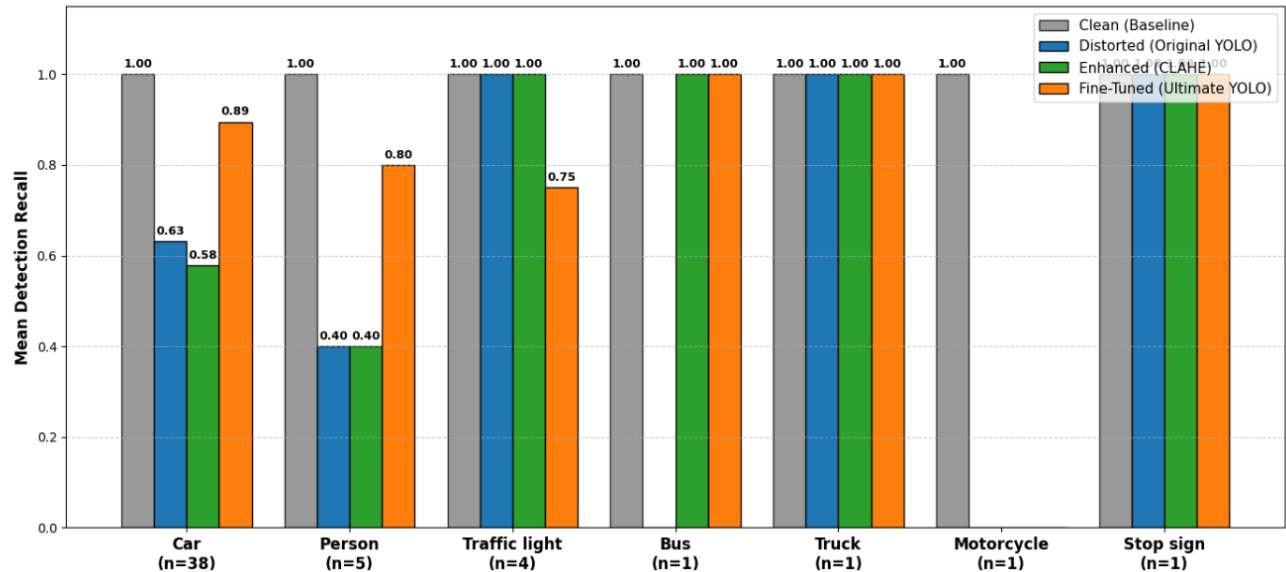


YOLOv8 results show a clear trade-off between clean accuracy and robustness

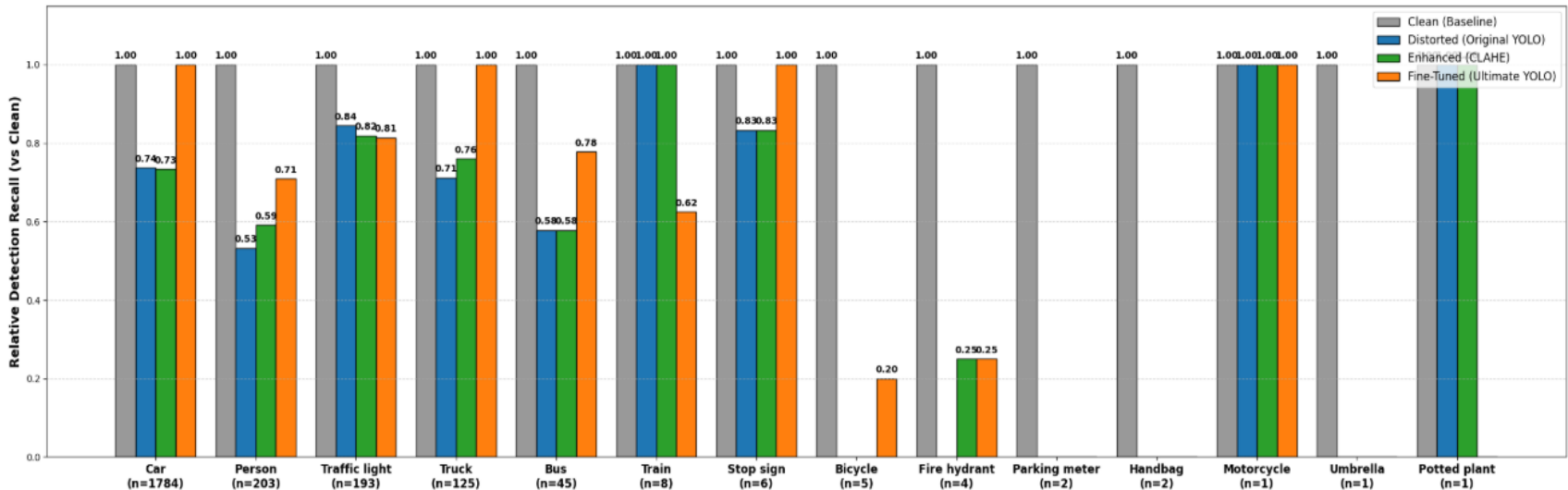
YOLOv8 Per-Class Recovery: Low Light

5 and 300-images sample

Per-Class Recall Recovery under High Severity Low-Light (b = -0.3)
Project by: Ayelet & Noa

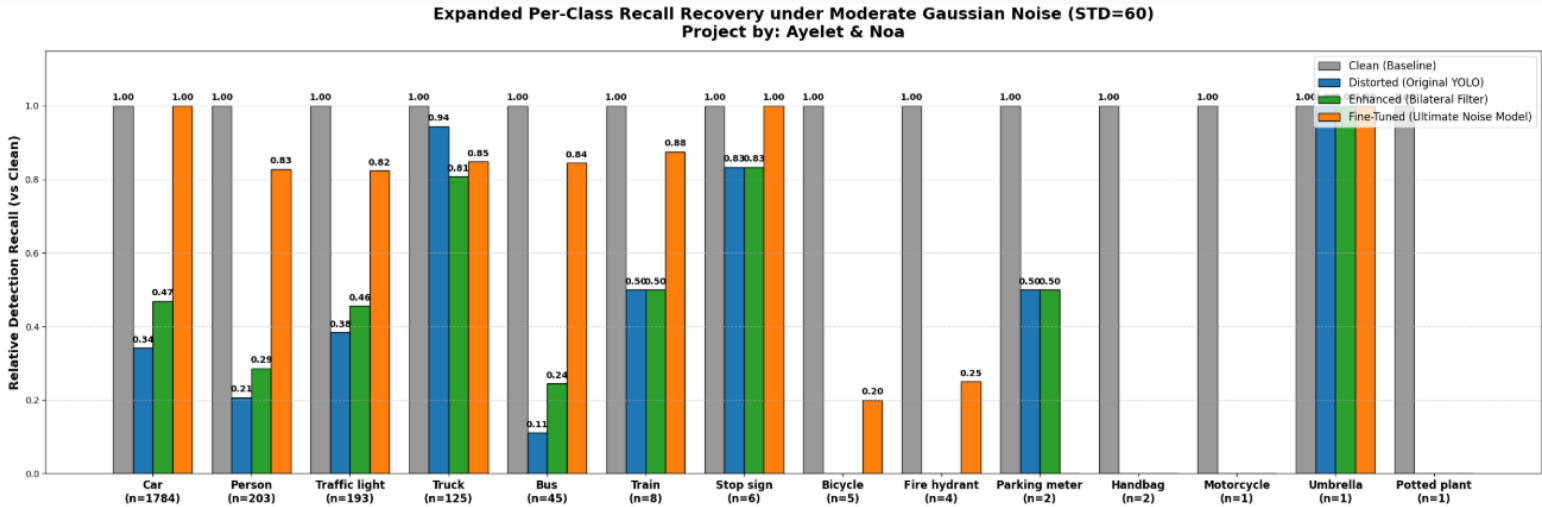
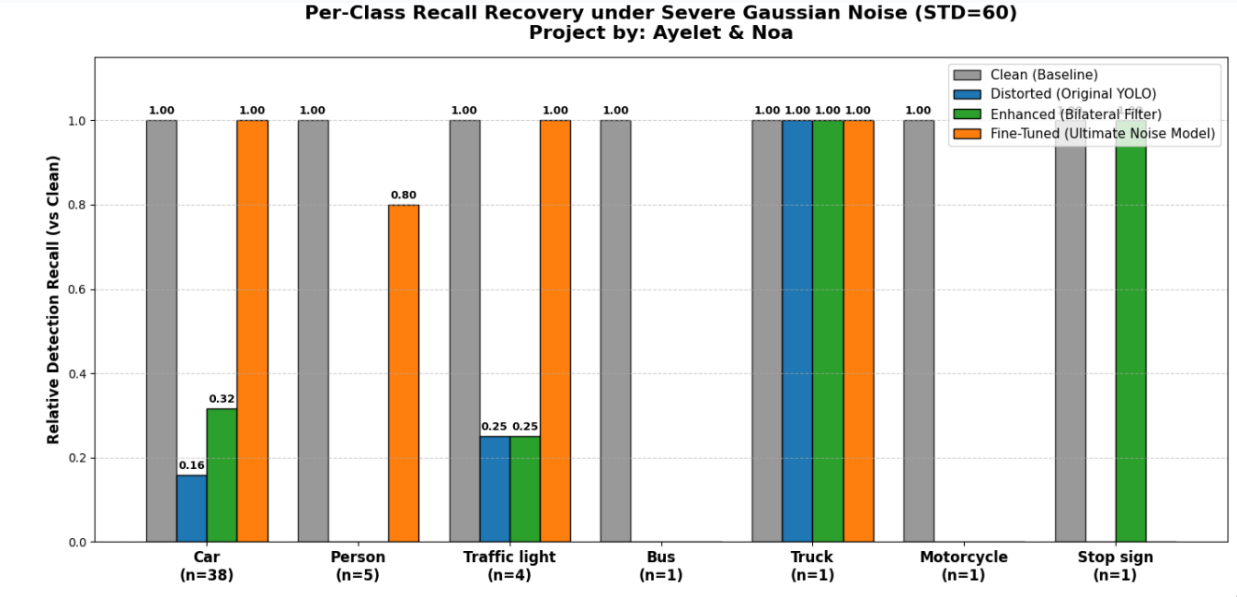


Expanded Per-Class Recall Recovery under Low-Light (b = -0.3)
Project by: Ayelet & Noa



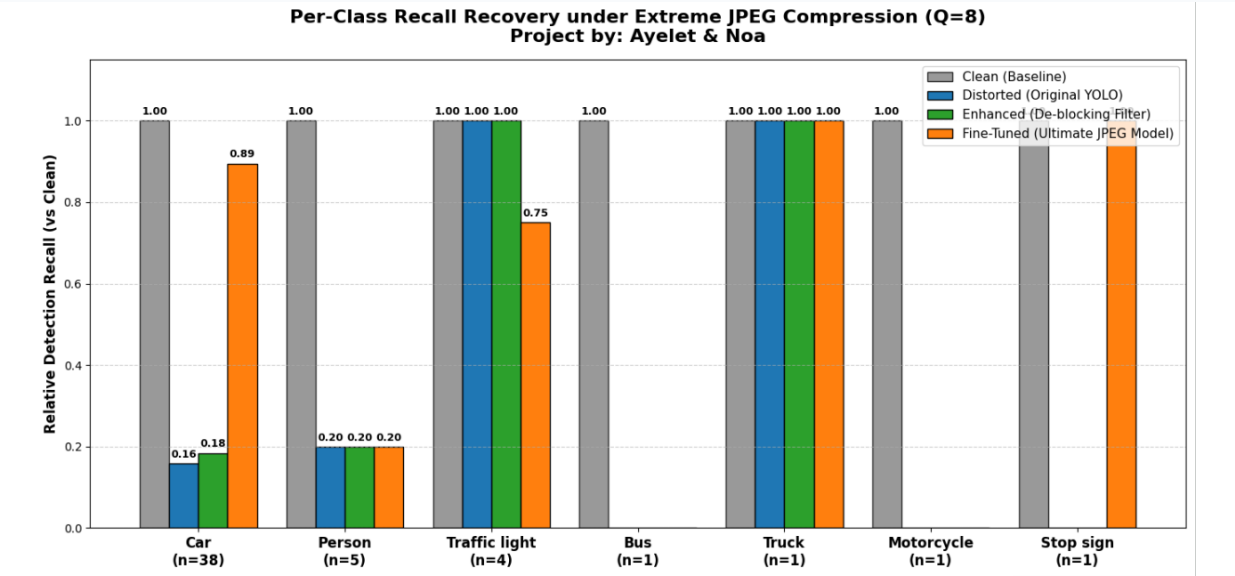
YOLOv8 Per-Class Recovery: Gaussian Noise

5 and 300-images sample

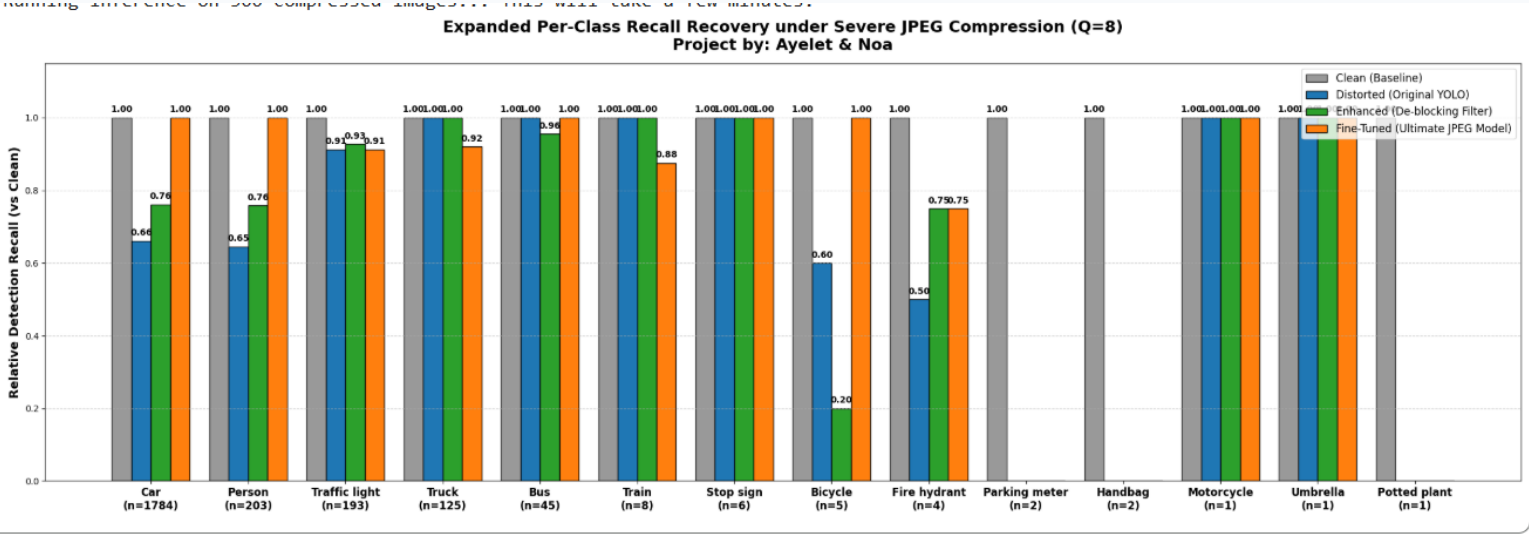


YOLOv8 Per-Class Recovery: JPEG Compression

5 and 300-images sample



Running inference on 300 compressed images will take a few minutes.



Mask R-CNN: Segmentation Robustness

Instance masks become unstable when textures are corrupted

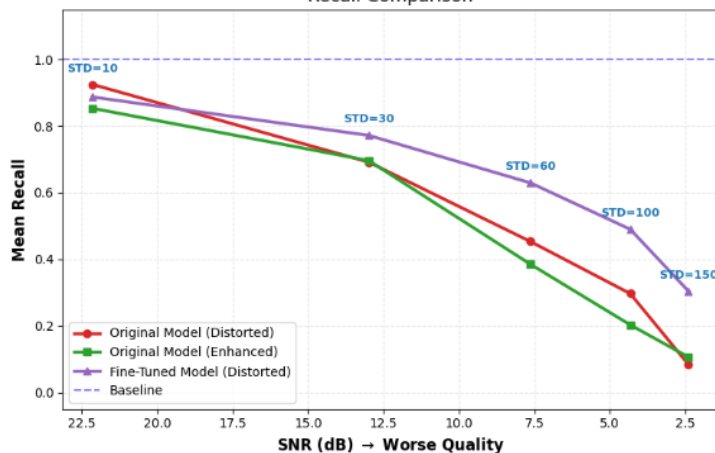
Classical filters can help, but the effect depends on the distortion

Fine-tuning was trained with pseudo-labels generated from clean images

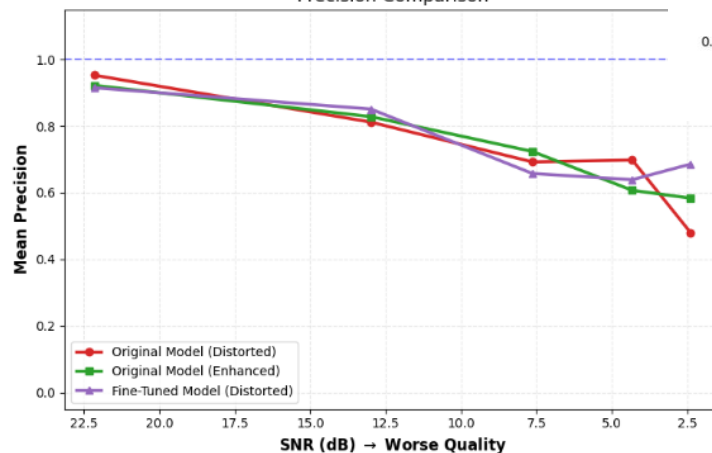
At extreme noise and compression, the fine-tuned model preserved higher recall

The Grand Finale: Gaussian Noise & Fine-Tuning Performance
Project by: Ayelet & Noa

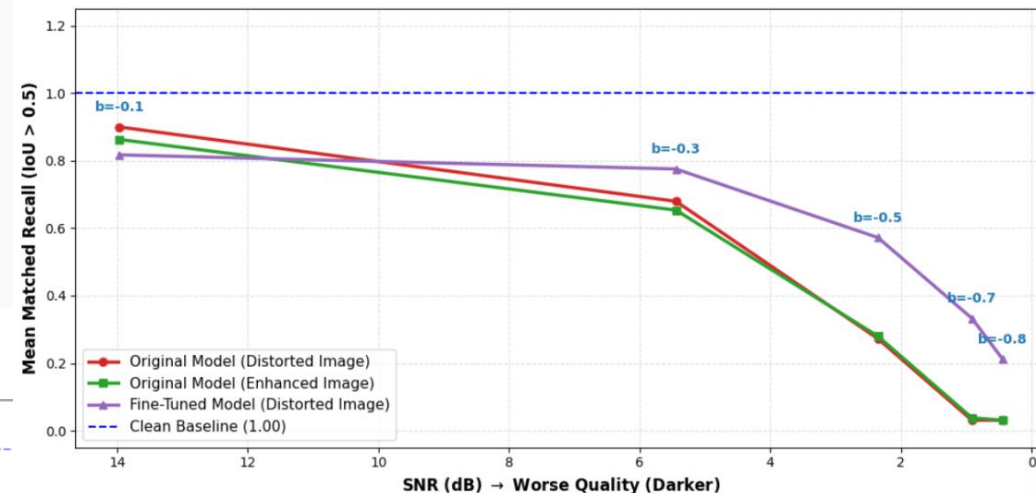
Recall Comparison



Precision Comparison



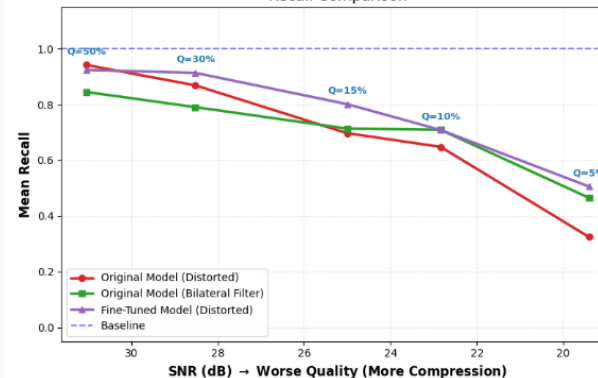
The Grand Finale: Low Light Recall Comparison
(Original vs. Mathematical Enhancement vs. Deep Fine-Tuning)



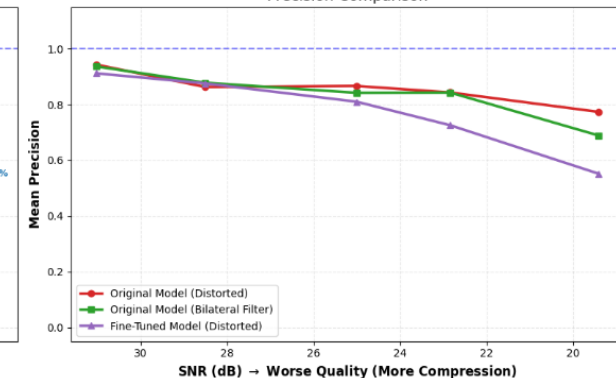
Fine-tuning compared against the original model and classical enhancement

The Grand Finale: JPEG Compression & Fine-Tuning Performance
Project by: Ayelet & Noa

Recall Comparison

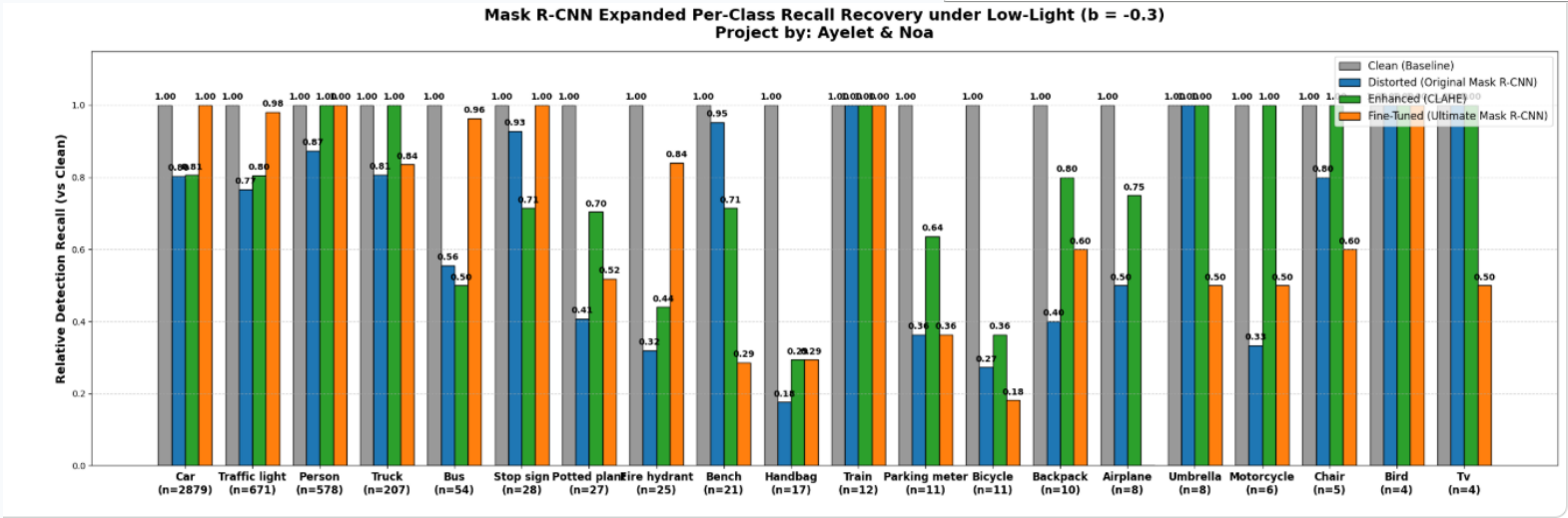
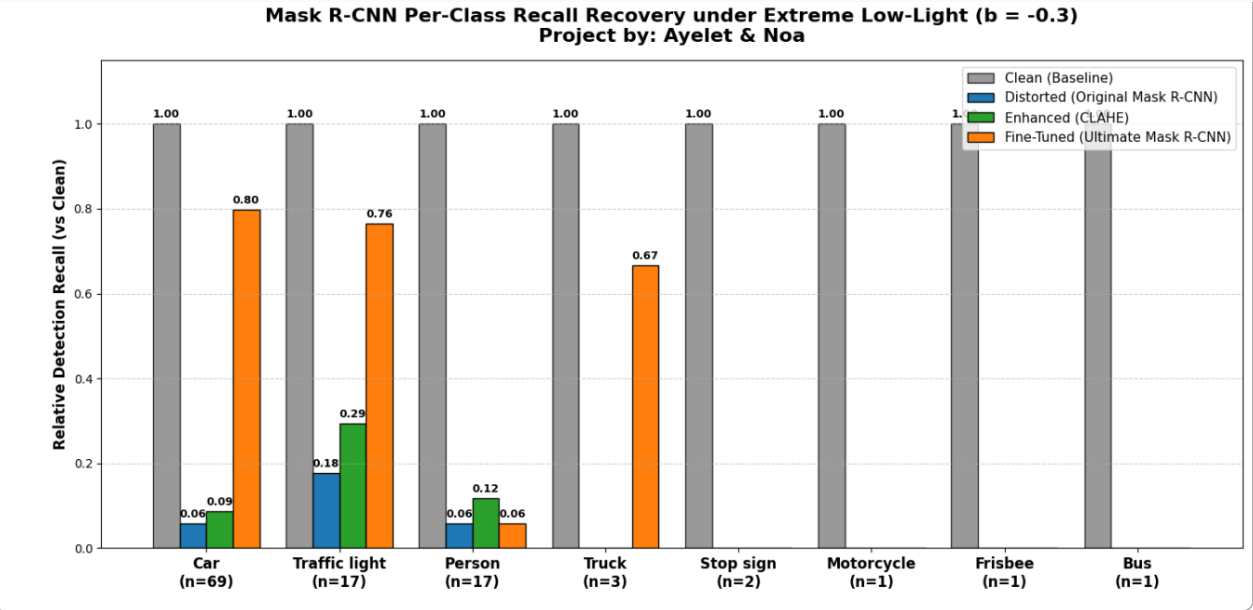


Precision Comparison



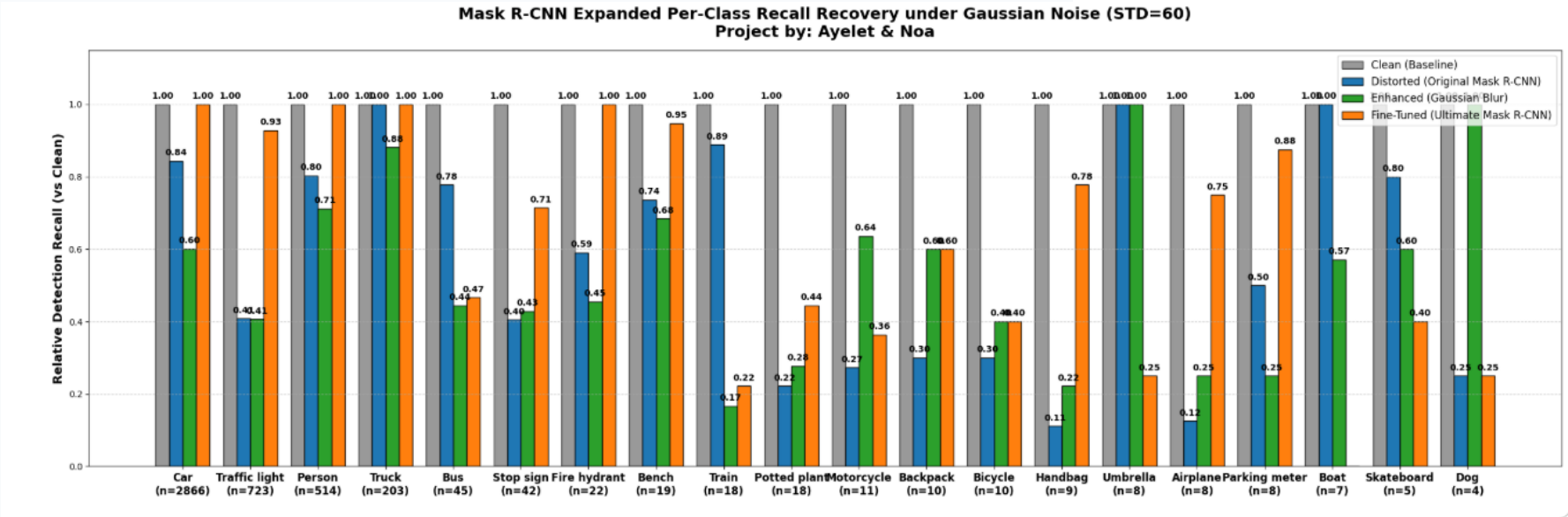
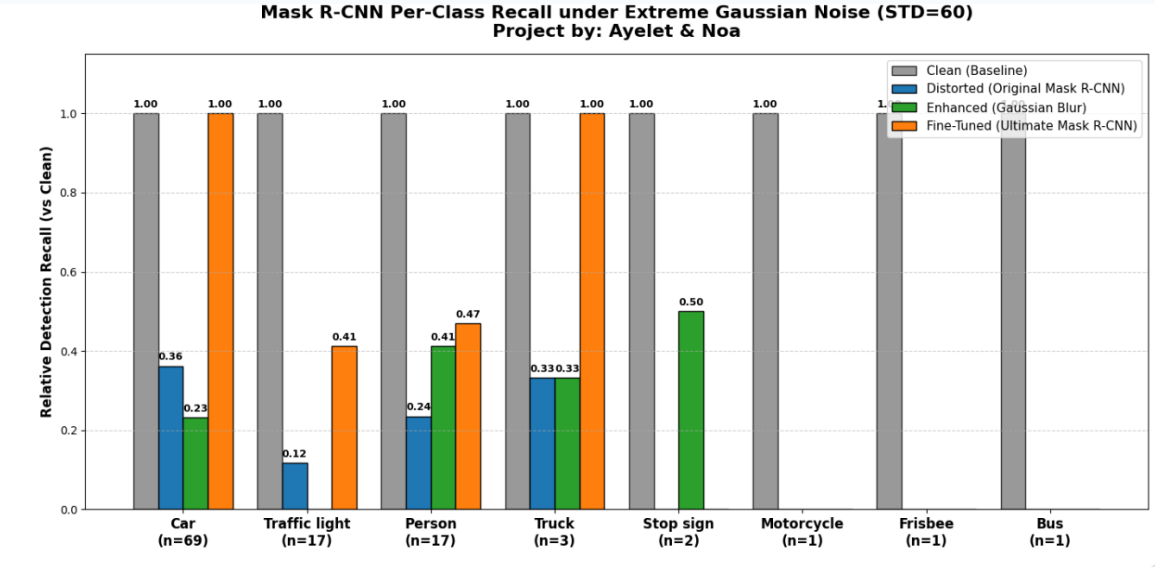
Mask R-CNN Per-Class Recovery: Low Light

5 and 300-images sample



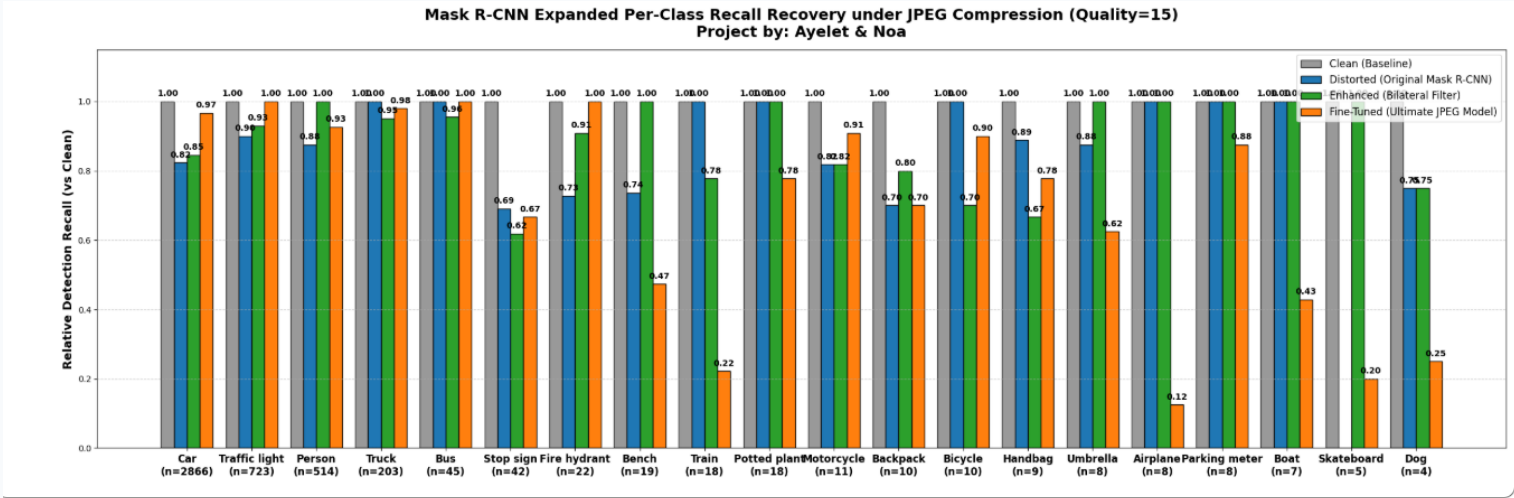
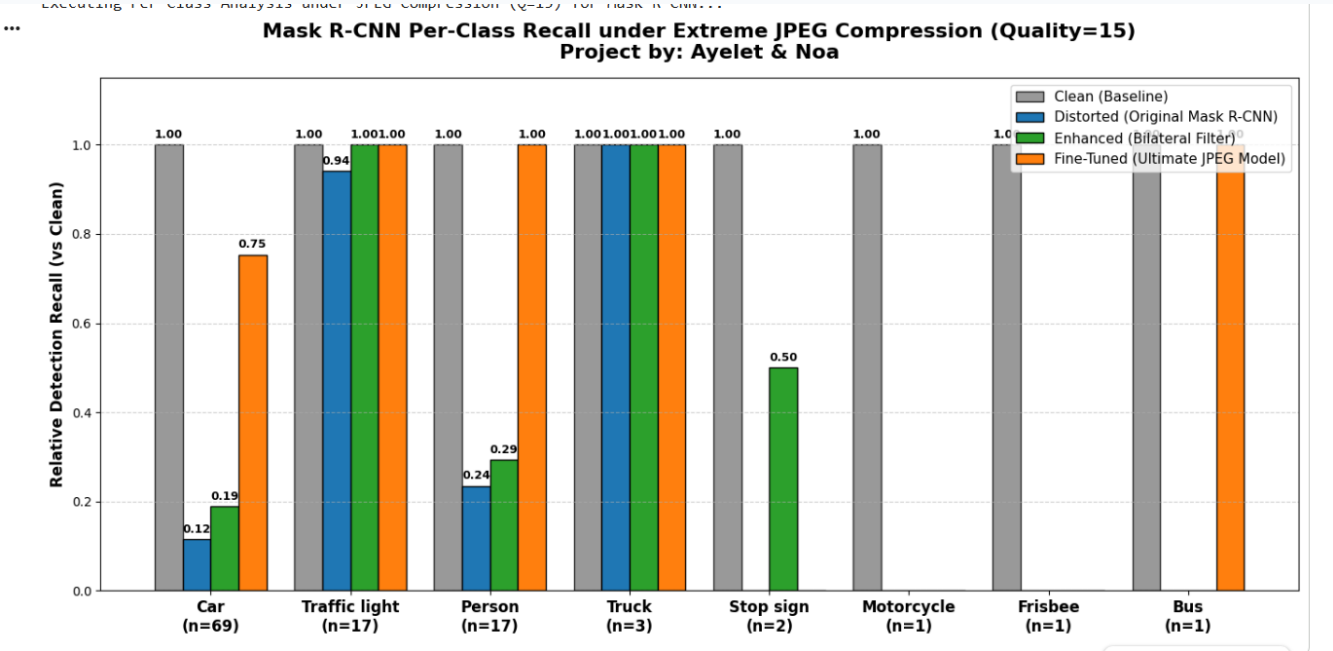
Mask R-CNN Per-Class Recovery: Gaussian Noise

5 and 300-images sample



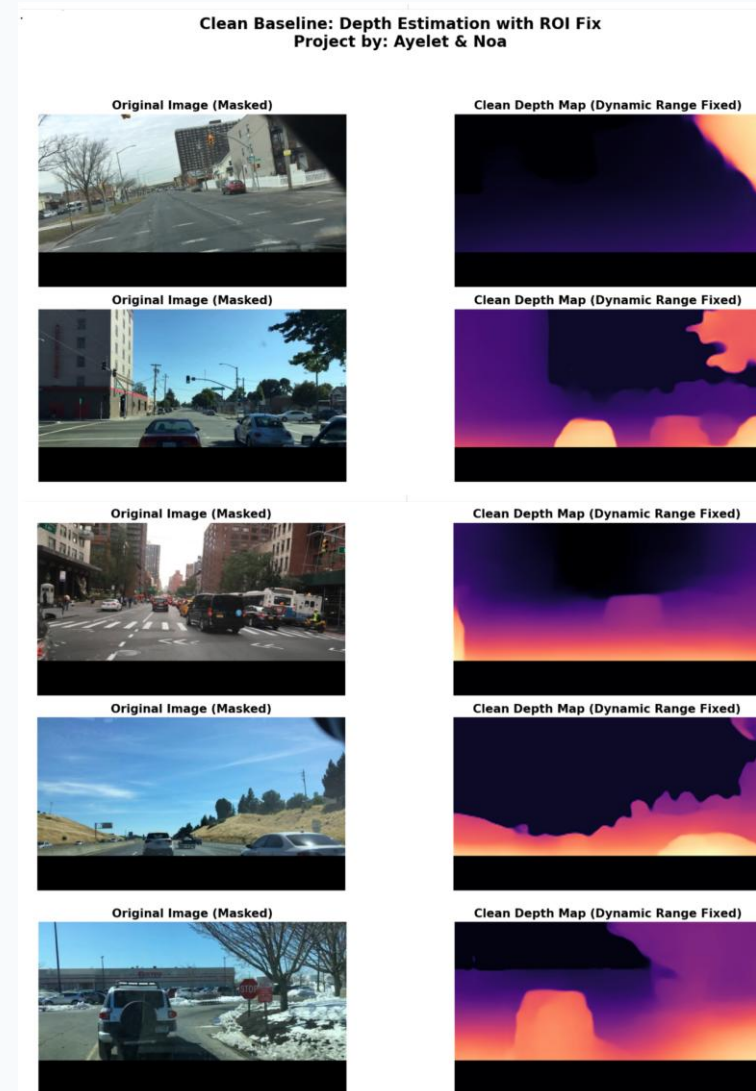
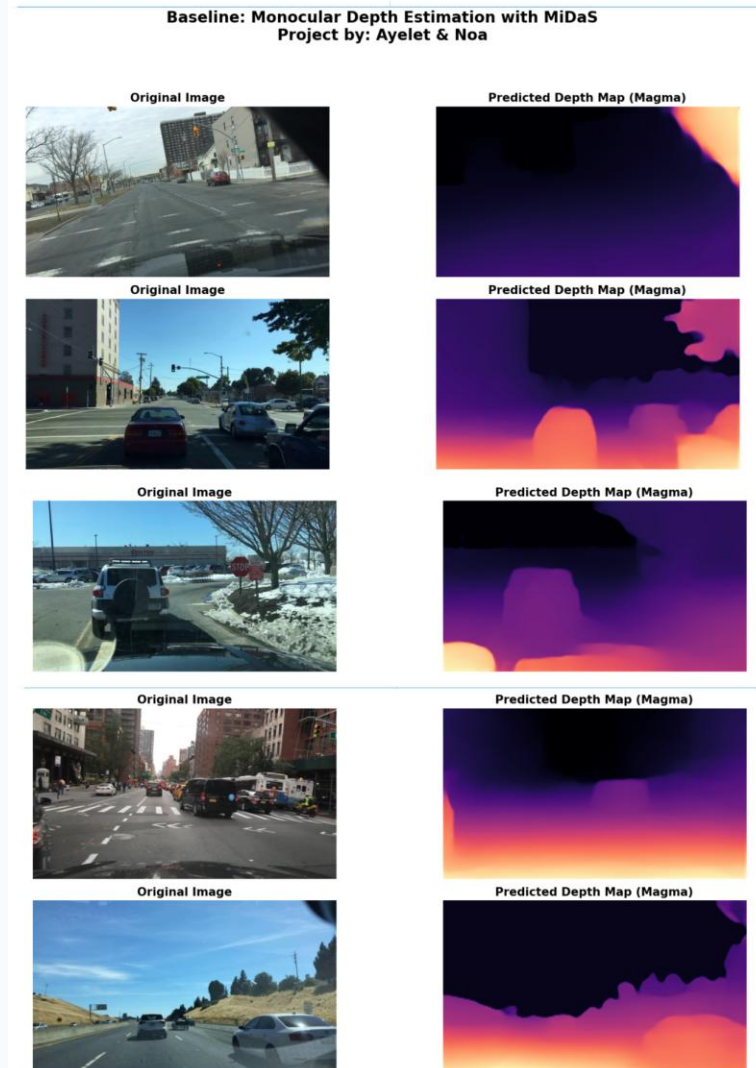
Mask R-CNN Per-Class Recovery: JPEG Compression

5 and 300-images sample



MiDaS: Depth Estimation Setup

MiDaS predicts a relative depth map from a single image
The dashboard region was removed from evaluation using an upper 80% ROI
Depth maps were normalized before comparison
RMSE captures numeric error; SSIM captures structural similarity



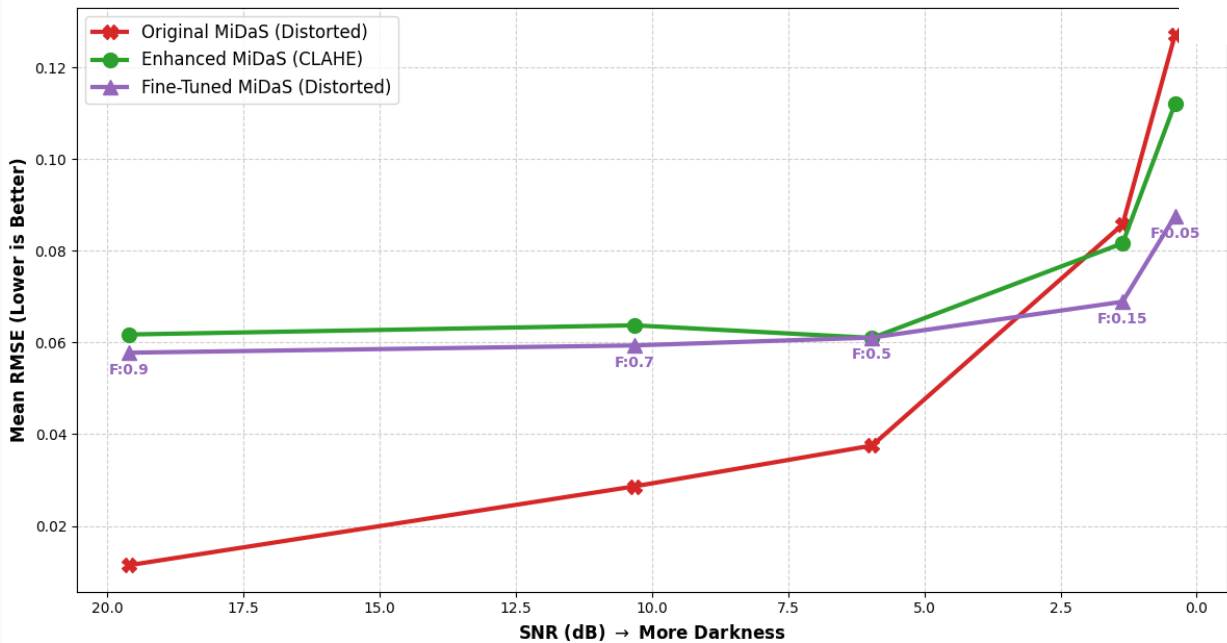
MiDaS under Low Light

Darkness weakens the visual signal used for depth estimation

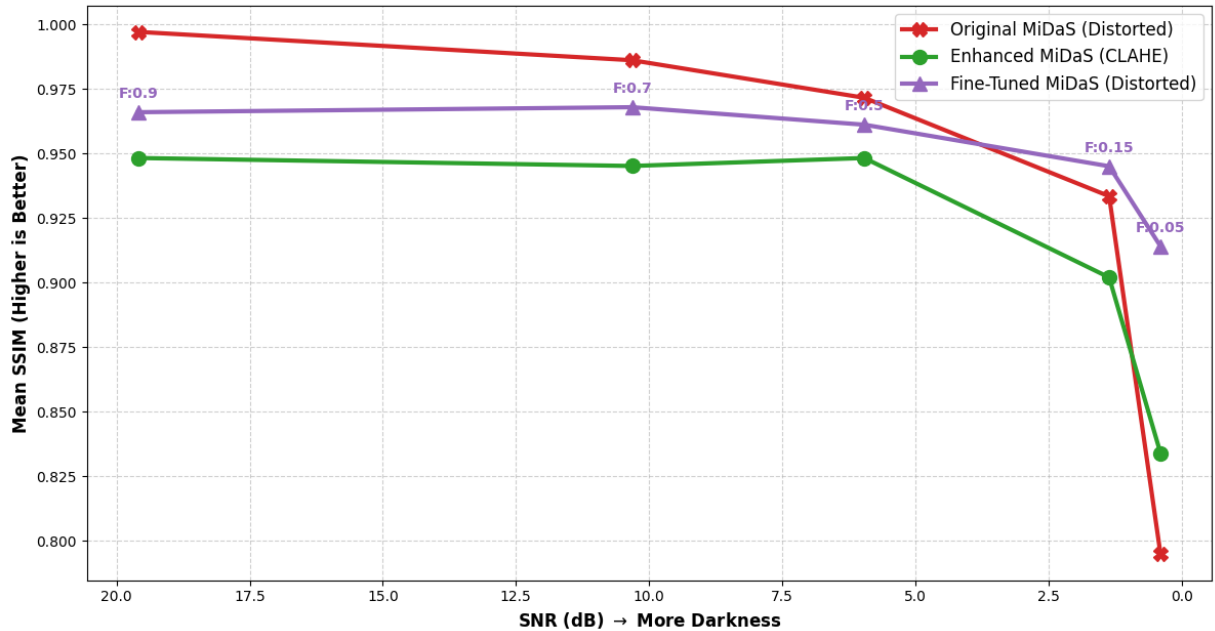
RMSE increases as the image becomes darker
SSIM decreases because the depth structure changes

CLAHE helps mainly under strong darkness
Fine-tuning gives the most reliable recovery at extreme low light

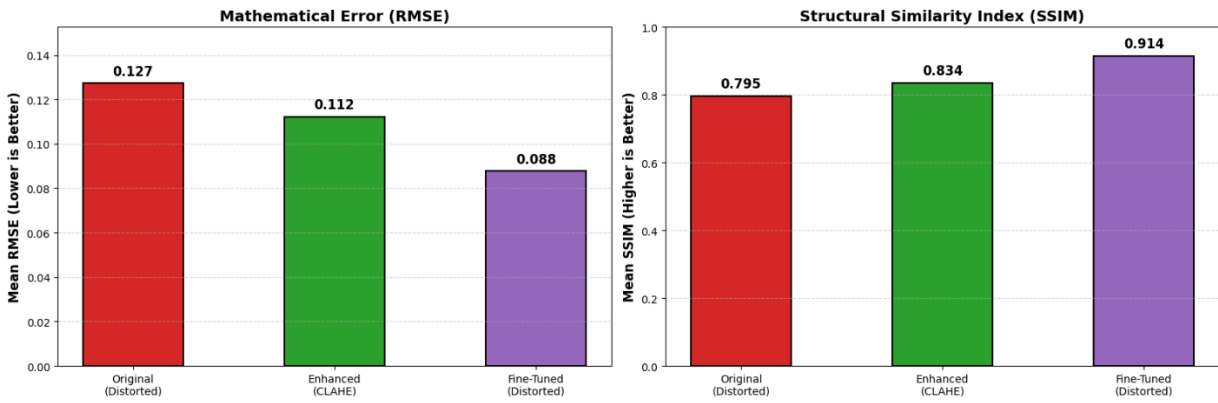
MiDaS Depth Perception: Original vs. Enhanced vs. Fine-Tuned
Project by: Ayelet & Noa



Structural Similarity Index (SSIM) under Low Light
Project by: Ayelet & Noa



MiDaS Performance Under Extreme Low Light (Factor: 0.05)
Project by: Ayelet & Noa



Fine-tuned MiDaS improves both RMSE and SSIM at the darkest level

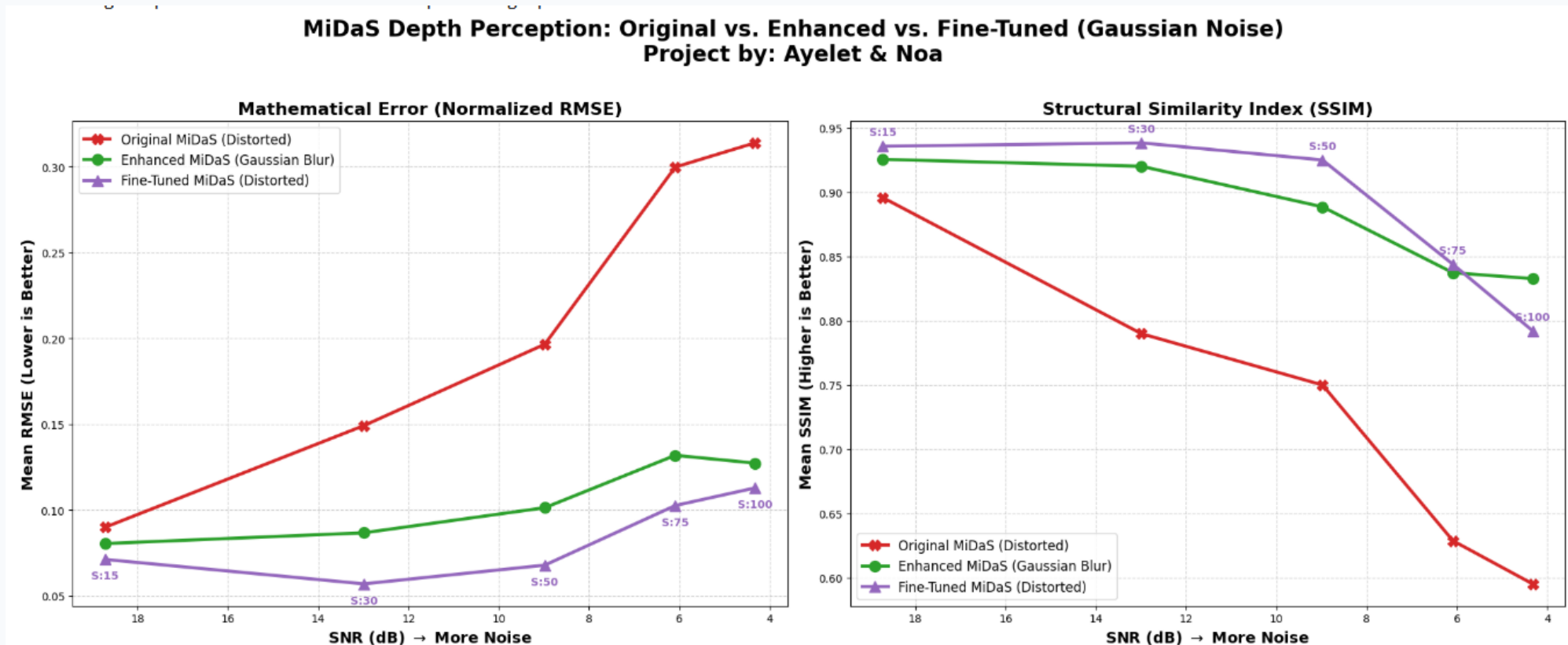
MiDaS under Gaussian Noise

The noisy baseline shows rising RMSE and falling SSIM

Gaussian Blur improves results at every noise level

Fine-tuning achieves the lowest RMSE across all tested noise levels

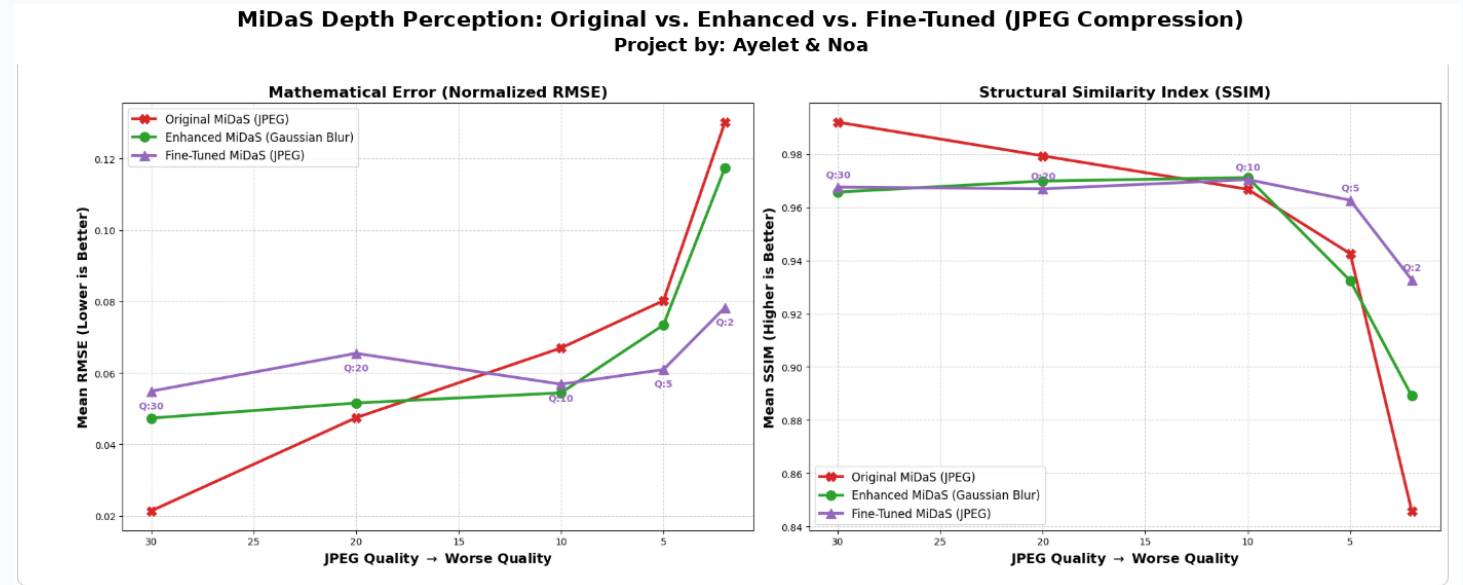
At very high noise, blur can retain slightly smoother structure



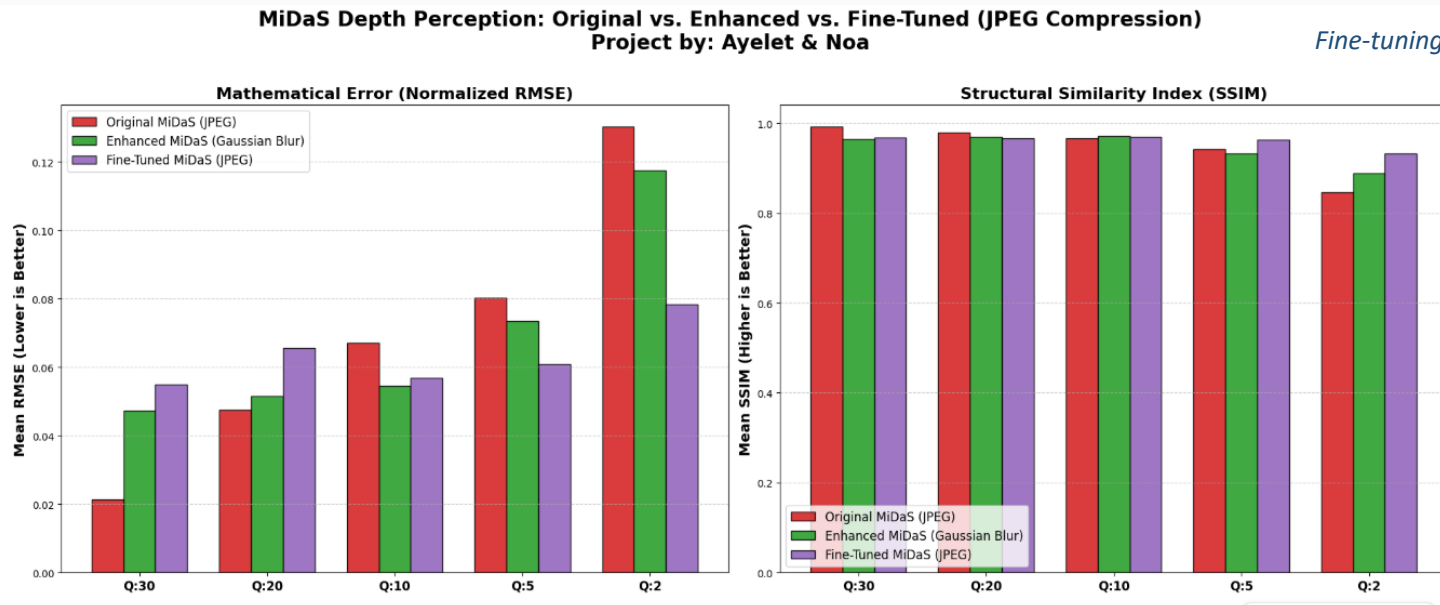
Gaussian Blur helps consistently; fine-tuning focuses on lowering numeric depth error

MiDaS under JPEG Compression

The original model is strongest when compression is mild
Gaussian Blur partially reduces blocking artifacts
Fine-tuning becomes clearly better at low quality factors
At Q=2, the fine-tuned model gives the lowest RMSE and highest SSIM



Fine-tuning trades a small loss at high quality for stronger robustness at severe compression



What We Learned

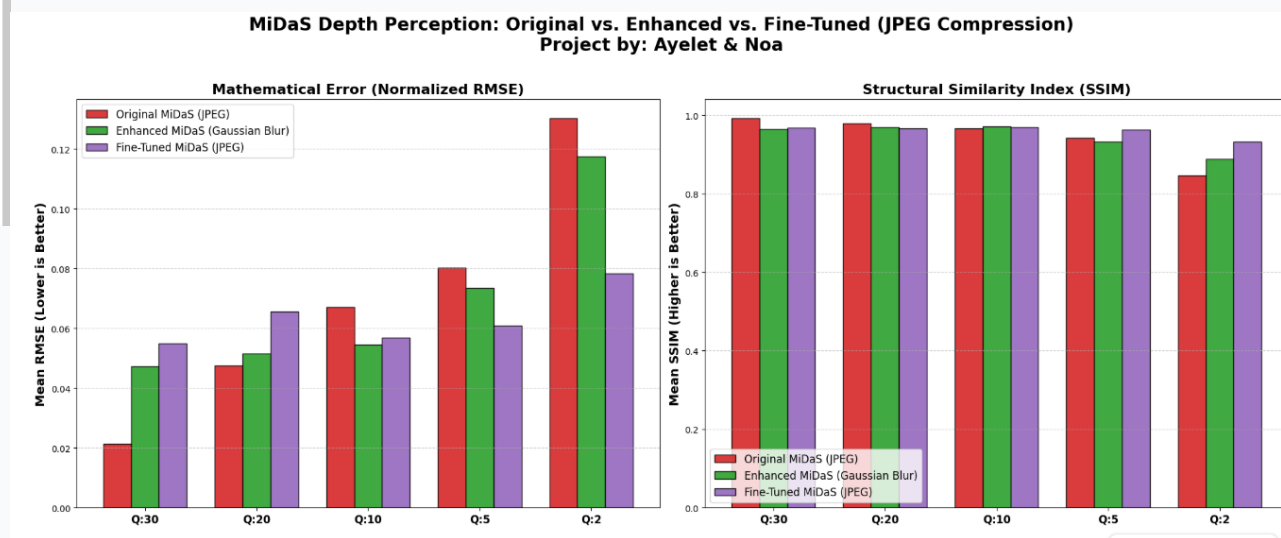
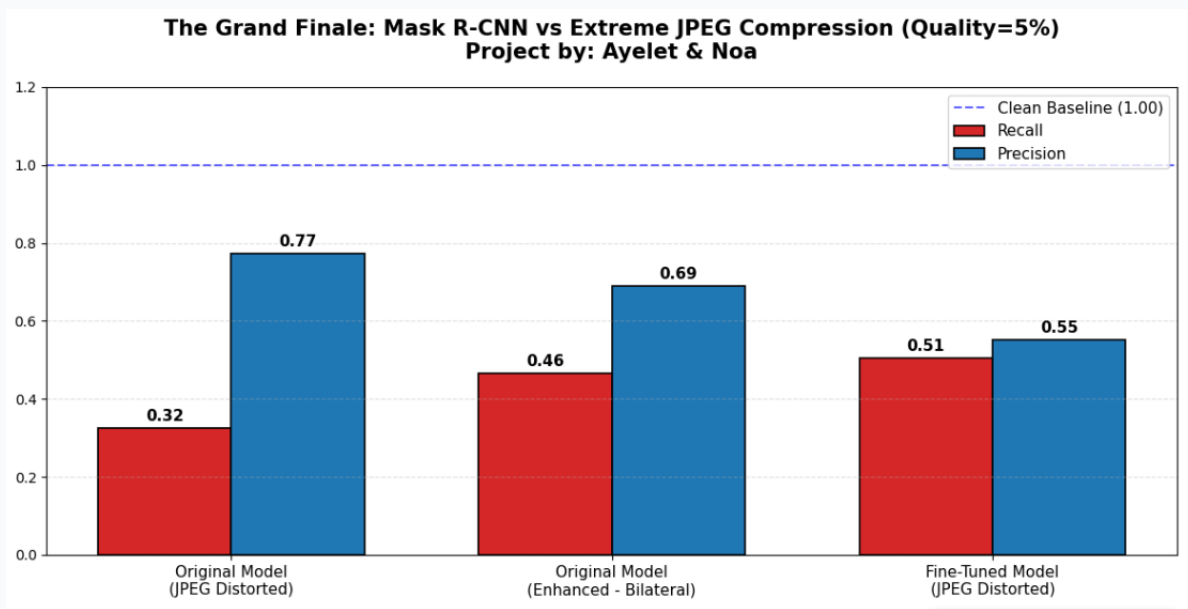
Model robustness depends strongly on the distortion type

Classical enhancement is useful when the distortion has a simple signal-processing structure

Fine-tuning is more reliable when distortions become severe or structural

A single metric is not enough: recall, precision, RMSE and SSIM can tell different stories

The best system should choose between original, enhanced and fine-tuned models according to image quality



Final results highlight the advantage of fine-tuning under severe degradation

Conclusion

Computer vision models are reliable on clean images, but performance drops under realistic distortions
Classical enhancement is simple and useful, especially for Gaussian noise
Fine-tuning provides stronger robustness under severe low light, noise and JPEG compression
Robust real-world systems should combine quality assessment, preprocessing and model adaptation

This course was interesting and we really enjoyed,

Thank you 😊

Ayelet & Noa

